

Anthropometric Profile of Indian Table Tennis Players

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ABSTRACT

The aim of this study was to determine anthropometric profile of Indian Table Tennis Players (male & female) who participated in the National Camp during 2017-18 at Sports Authority of India, NS NIS Patiala. A total of 32 players (19 male and 13 female) aged between 20 to 30 years were examined for this study. Various anthropometric measurements viz. height, weight, diameters (humerus, wrist, femur and ankle), Circumferences (upper arm normal & flex, forearm, thigh and calf), skinfolds (biceps, triceps, forearm, subscapular, suprailiac, supraspinale, thigh and calf) were taken following standard techniques. Mean, SD were calculated and t test was conducted for comparison. The mean value of decimal age, height, weight and height-weight ratio were 23.47 years, 168.68 cm, 65.52 kg and 41.97 for male respectively. For female the mean decimal age, height, weight and height-weight ratio were 24.59 years, 157.83 cm, 58.86 kg and 40.67, respectively. Mean somatotype of male was 3.12 - 5.39 - 2.19, whereas for female it was 5.71 - 4.63 - 1.28. Mean of body fat percentage was 16.36 and 30.49 for male and female, respectively. It had been observed that body fat content in female players was higher than male players.

Keywords

Anthropometry, Height, weight, somatotype, body composition

INTRODUCTION

The interest in anthropometric characteristics, body composition and somatotype from different competitive sports has increased over the last decades. Physique, body composition, physical growth and one's motor development are of fundamental importance in developing the criteria of talent selection and development in

sports (Sodhi, 1991). Anthropometric profile may indicate whether a player would be suitable to participate at the highest level in a specific sport (Bourgois et al., 2000).

Several studies have assessed the anthropometric, physiological and motor skill attributes of individual sports (Bencke et al., 2002; Thelwell et al., 2006). Table Tennis is a dynamic sport

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characterized by highly developed motor skills such as agility, reaction speed, explosive power, strength, and eye movement and its coordination (Rodrigues, 2002; Jafarzadehpur, 2004 and Kondric, 2007). No doubt physical training plays an important role to reach sport success, but we can never neglect the fact that, at the same training level, the best performances are shown by those players who have better compatible anatomic conditions (Kondrič et al., 2013). The importance of early sports talents detection, biotype influence in sport performance and lack of definition of suitable morphology of Table Tennis players, it is necessary to do analysis that may provide specific anthropometric reference for this sport (Carrasco et al., 2010, Munivrana et al., 2011). In India, Anthropometric studies on top level table tennis players are still lacking. Keeping this in view the present study has been conducted to make a base line data for table tennis players.

METHODOLOGY

For the present study anthropometric data has been collected from 32 National Table tennis players (19 male and 13 female) who participated in the National camp 2017-18 at NS NIS Patiala. The selected anthropometric parameters for this research are height, weight, diameters of humerus, wrist, knee & ankle; circumference of upper arm normal, upper arm flex, forearm, thigh and calf; skinfolds of biceps, triceps, forearm,

sub-scapular, supra-iliac, supra-spinale, thigh and calf. Anthropometric Rod (GPM), Digital Weighing Machine, Bicondylar Vernier calliper (Holtain), Measuring Tape (Hoechstmass) and Skinfold Caliper (GPM) were used for taking body measurements. Anthropometric measurements were taken on right side of the body by following standard techniques of Ross et al. (1978). Data were analysed for somatotype and body composition. The formula to calculate height weight ratio (HWR) is mentioned as below:

$$\text{HWR} = \text{Height (cm)} / \text{Cube root of Weight (kg)}$$

Somatotype components (endomorph, mesomorph and ectomorph) of the subject's were computed according to equations of Carter (1980). Bone mass and Muscle mass has been calculated through Matiegka's methods (1921). Body fat percentage was calculated from skinfolds of biceps, triceps, sub-scapular and supra-iliac using equations of Durnin and Womersley (1974) for density. Body density was converted into body fat according to equations of Brozek et al. (1963).

$$\% \text{ Body Fat} = [(4.57 \div \text{Body Density}) - 4.142] \times 100$$

For plotting of somatotypes on somatochart 'X' and 'Y' coordinates were calculated as follows:

X = Ectomorphic component – endomorphic component

Y = 2 x mesomorphic component – (endomorph component + ectomorph component).

Mean values, Standard deviation (SD) and Ranges were calculated. Further, t-test was conducted for the comparison of height and weight of both genders

between the data of present study and data of Europe 1973 (Eiben & Eiben 1979).

RESULTS & DISCUSSION

Table-1:
Overall descriptive statistics of anthropometric Parameters for Indian Table Tennis Players (N=32).

Anthropometric Parameters	Male (N=19)				Female (N=13)			
	Mean	SD	Min.	Max.	Mean	SD	Min.	Max.
Decimal Age (years)	23.47	3.37	17.99	30.54	24.59	4.73	18.53	33.63
Height (cm)	168.68	5.25	161.2	187.3	157.83	4.75	147.60	163.90
Weight (kg)	65.52	7.91	48.4	76.2	58.86	7.05	49.60	70.20
Height-weight ratio	41.97	1.99	39.05	45.89	40.67	1.43	37.62	43.27
Endomorphy	3.12	1.21	1.25	5.48	5.71	1.00	4.16	7.76
Mesomorphy	5.39	1.26	2.74	7.41	4.63	1.16	3.12	7.18
Ectomorphy	2.19	1.39	0.34	5.01	1.28	0.88	0.10	3.09
Bone Mass (%)	15.90	1.42	13.53	19.47	13.37	0.74	12.14	14.94
Muscle Mass (%)	50.04	3.75	42.94	56.79	39.63	2.61	33.43	43.52
Body Fat (%)	16.36	4.43	8.55	23.35	30.49	3.09	26.07	35.46

Overall descriptive statistics of anthropometric parameters of Indian table tennis male and female players is shown in table 1. The mean value of decimal age for male players was 23.47 years, whereas female it was 24.59 years. Mean height and weight were 168.68 cm, 65.52 kg and 157.83 cm, 58.86 kg for male and female respectively. The male tennis players have 1.30 unit higher height weight ratios compared to female counterparts. Thus male players were proportionally lighter than female tennis players. The

mean value of the somatotype components (endomorph - mesomorphy - ectomorphy) were 3.12 - 5.39 - 2.19 and 5.71 - 4.63 - 1.28 for male and female respectively. The mean somatotype of male players on somatochart falls at endomorphic-mesomorph sector (3.12 - 5.39 - 2.19), and female at mesomorphic-endomorph sector (5.71 - 4.63 - 1.28). The mean bone mass %, muscle mass % and body fat % of male players were 15.90 %, 50.04 %, and 16.36 %, respectively. For female the mean values of bone mass %,

muscle mass % and body fat % were 13.37 %, 39.63 % and 30.49 %, respectively.

According to Eiben & Eiben, 1979 report (Europe, 1973), the mean age was 23.5 years for male and 21.2 years for female, mean height-weight ratio was 42.3 for male and 42.5 for female, the mean somatotype of male was 3.5 – 3.9 – 2.5 and for female it was 4.5 – 3.3 – 2.7. Holzer et al. (1984) showed a wide distribution of somatotypes ($3\frac{1}{2}$ - 4 - $2\frac{1}{2}$) among Australian male table tennis players, similar to that of other European players (Eiben & Eiben, 1979). European female table tennis

players were meso-endomorphs.

Table 2 (a) and 2 (b), shows the comparison of height and weight of present study with the Europe, 1973 (Eiben & Eiben, 1979) of male and female, respectively. It has been found that there is statistical significant difference ($t=3.25$, $p=0.0018$) in height and no significant different in weight of males (table 2 a). Similarly, for females statistical significant difference ($t = 2.71$, $p = 0.0096$) in height and non-significant difference in weight has been observed (table 2 b). It signifies that European players are taller in height compare to Indian players.

Table 2 (a):
Comparison of Height & Weight of Present study with European Male Players, 1973 (Eiben & Eiben 1979)

Anthropometric Parameters	Present Study (N=19)	Europe, 1973 (N=48)	T	p value
Height (cm)	168.68 $\pm$ 5.25	174.2 $\pm$ 6.6	3.25*	0.0018
Weight (kg)	65.52 $\pm$ 7.91	69.7 $\pm$ 8.5	1.84	0.069

Table 2 (b):
Comparison of Height & Weight of Present study with European Female Players, 1973 (Eiben & Eiben 1979)

Anthropometric Parameters	Present Study (N=13)	Europe, 1973 (N=31)	T	p value
Height (cm)	157.83 $\pm$ 4.75	163.6 $\pm$ 7	2.71*	0.0096
Weight (kg)	58.86 $\pm$ 7.05	56.9 $\pm$ 7.4	0.81	0.4212

Fig. 1, 2 & 3 shows the bar diagrammatic representation of anthropometric parameters, somatotype and body

composition for both male and female players of Indian table tennis respectively.

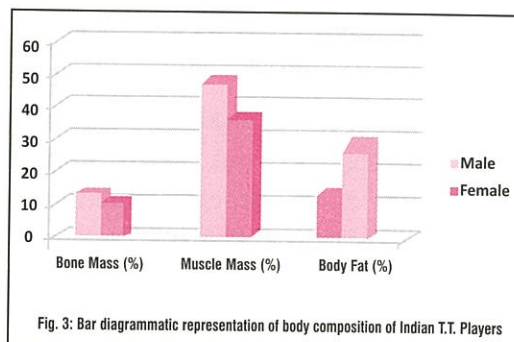
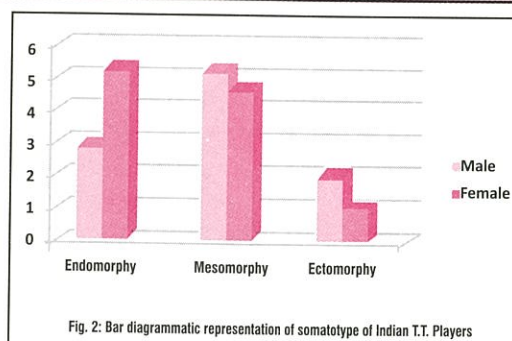
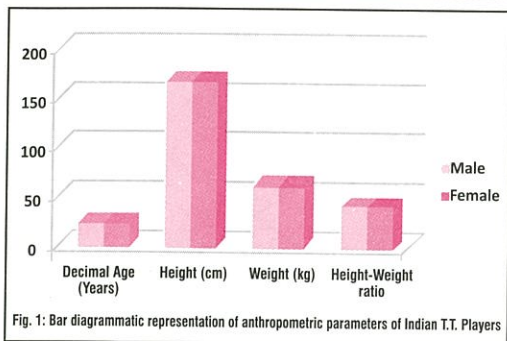


Fig. 4 & 5 depicts the pie diagrammatic representation of frequency of somatotype classification of Indian table tennis players for male and female respectively. It denoted that endomorphic - mesomorph frequency

(53%) is highest followed by balance mesomorph (21%) in male players while mesomorphic - endomorph with highest frequency (62%) followed by mesomorph - endomorph (38%) is observed in female players.

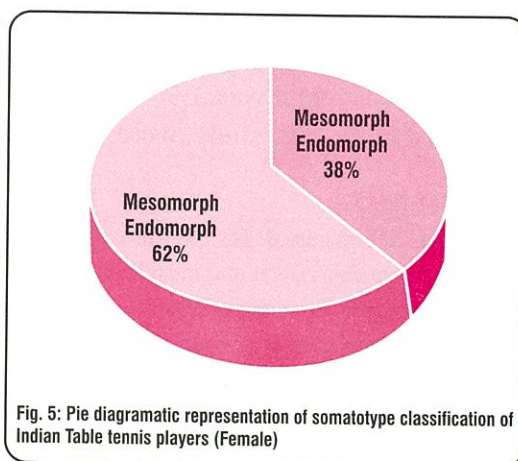
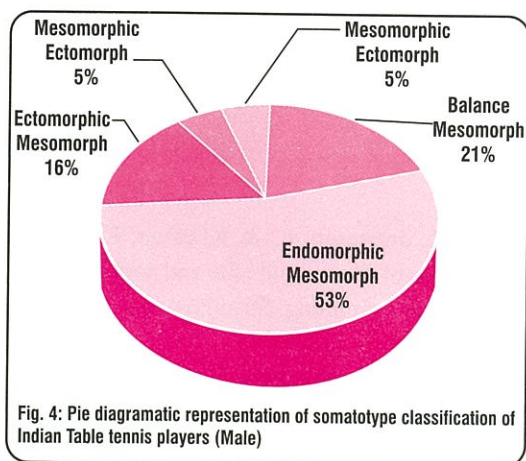
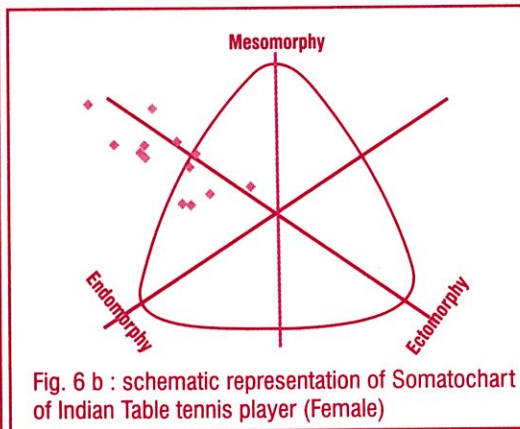
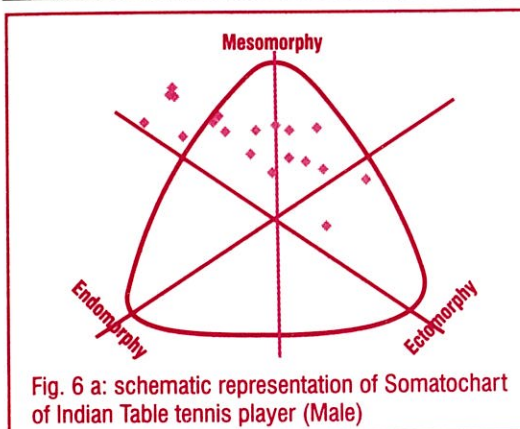


Fig. 6 a & 6 b, shows the schematic diagrammatic representation of

somatochart of Indian table tennis male and female players respectively.



CONCLUSION

It has been observed that there is statistical significant difference in height of male ($t = 3.25$, $p = 0.0018$) and female player ($t = 2.71$, $p = 0.0096$), non-significant difference in weight of males and females when comparing the present study with Eiben & Eiben, 1979 report (Europe 1973). It signifies the European players are taller in height compare to Indian players.

The mean somatotype of male tennis players in the present study was $3.12 - 5.39 - 2.19$ whereas for European players it was $3.5 - 3.9 - 2.5$ and for Australian players it was $3\frac{1}{2} - 4\frac{1}{2} - 2\frac{1}{2}$. The players in this study were found

similar in endomorphy and ectomorphy but the mesomorphy was more than above studied players.

The female tennis players in this study ($5.71 - 4.63 - 1.28$) was more endomorphic and mesomorphic but lesser ectomorphic than European female players ($4.5 - 3.3 - 2.7$). The female players in this study were proportionally heavier for height than European female players

The average percent body fat of female players was 30.49%. They were advised to reduce fat and not to increase further so that fitness can be kept at optimal level.

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